

A Differentiable Brain Tumor Mass Effect Solver

Marco L. Wittrich^{1,2} Michal Balcerak⁴ Lucas Zimmer¹ Ivan Ezhov³ Björn Menze⁴ Marie-Christin Metz¹
Nicole Tueni⁵ Benedikt Wiestler^{1,2} Jonas Weidner^{1,2}

¹ AI for Image-Guided Diagnosis and Therapy, TUM ² Munich Center for Machine Learning (MCML)
³ Chair for AI in Healthcare and Medicine, TUM ⁴ University of Zurich ⁵ Continuum Mechanics and Biomechanics, FAU Erlangen-Nürnberg

Motivation

- A glioblastoma does not only infiltrate tissue. It also pushes that tissue out of the way.
- This **mass effect** deforms healthy brain, which in turn affects prognosis and where radiotherapy targets are drawn.
- The models we already have solve PDEs. They are accurate, but they are slow, difficult to implement, and you cannot differentiate through them.

Methods

- We lay a regular grid over the brain. Every node is a point mass, joined by springs to the 26 neighbors in its $3 \times 3 \times 3$ cube.
- The tumor concentration, which comes from a separate Fisher-Kolmogorov growth fit [3], acts as a potential that pushes nodes outward. The spring lattice, weighted by tissue stiffness, holds them back by penalising any change from the rest length r_{ij}^0 . A fixed shell of boundary nodes stands in for the skull.

$$E(\mathbf{u}) = \underbrace{\alpha \sum_{i \in \mathcal{F}} c(\mathbf{x}_i + \mathbf{u}_i)}_{\text{tumor potential}} + \underbrace{\beta \sum_{(i,j) \in \mathcal{E}} k_{ij} \left(\|\mathbf{x}_i + \mathbf{u}_i - \mathbf{x}_j + \mathbf{u}_j\| - r_{ij}^0 \right)^2}_{\text{tissue-weighted spring lattice}} \quad (1)$$

- Stiffness values come from elastography [5]. Gray matter is $2.7\times$ stiffer than white matter, and CSF is $10\times$ softer.
- We minimize the energy by gradient descent, taking gradients from **JAX**. That is one forward run per patient, with nothing to invert.
- There is a **single free parameter**, $\alpha = 1000$, which we fixed once by grid search.

Evaluation

We start from 136 BraTS2021 patients [6] in SRI24 atlas space. Two GLIA inversions failed, so every method is scored on the same **134**. Each method produces a displacement field $\mathbf{u}$, which we apply to the shared healthy atlas labels A with the same nearest neighbor pull warp,

$$W(\mathbf{x}) = A(\mathbf{x} - \mathbf{u}(\mathbf{x})), \quad (2)$$

and compare the result against the patient tissue segmentation G (CSF, GM, WM, VT). Our main metric is per tissue $\Delta\text{Dice} = \text{Dice}(W) - \text{Dice}(A)$. On the voxels where atlas and patient disagree, $\mathcal{N} = \{\mathbf{x} : A \neq G\}$, we also report

$$\text{recall} = \frac{|\{\mathbf{x} \in \mathcal{N} : W = G\}|}{|\mathcal{N}|}, \quad (3)$$

$$\text{net} = \frac{|\{A \neq G, W = G\}| - |\{A = G, W \neq G\}|}{|\mathcal{N}|}. \quad (4)$$

We compute scores in peritumoral shells of 5, 10, 15 and 20 mm, and over the whole brain. The tumor itself is always left out.

Baselines

- **GLIA** inverts three parameters per patient and models ventricle compression explicitly [1].
- **FLE** solves linear elasticity forward once for a fixed tumor [2].
- **ANTs SyN** optimizes anatomical overlap directly instead of modelling any physics, so we read it as an upper bound rather than a competitor [4].

Results

Table 1. Model performance. Overlap improvement at 10 mm and over the whole brain (WB). Mean $\pm$ SEM, $n = 134$, $\times 10^{-2}$.

Model	$\Delta\text{Dice} \uparrow$		net $\uparrow$	
	10 mm	WB	10 mm	WB
SyN (ceiling)	8.05 $\pm$ 0.22	11.07 $\pm$ 0.16	17.30 $\pm$ 0.50	23.72 $\pm$ 0.33
PSM (ours)	2.36 $\pm$ 0.14	0.53 $\pm$ 0.04	4.87 $\pm$ 0.29	1.02 $\pm$ 0.07
FLE	0.44 $\pm$ 0.04	0.06 $\pm$ 0.01	0.87 $\pm$ 0.08	0.11 $\pm$ 0.01
GLIA	<u>0.88</u> $\pm$ 0.11	<u>0.52</u> $\pm$ 0.05	<u>1.07</u> $\pm$ 0.19	1.13 $\pm$ 0.09

Table 2. Deformation field characteristics, whole brain with the tumor masked out. Fold counts voxels with $\det J \leq 0$.

Model	$ \bar{u} $ (mm)	Fold (%) $\downarrow$	Smooth. $\downarrow$	Vort. $\downarrow$
SyN (ceiling)	2.833	0.200	0.1345	0.458
PSM (ours)	0.609	0.002	0.1267	0.477
FLE	0.119	0.000	0.1052	0.479
GLIA	0.418	0.000	0.0893	0.570

Overlap. PSM comes out ahead on both ΔDice metrics and on net reclassification at 10 mm. It is best or tied for CSF, GM and WM, and second behind GLIA for the ventricles.

Plausibility. Folding is negligible for all of the biophysical models, so the fields they predict stay physically sensible.

Cost. About 15 seconds per patient, against roughly 24 minutes for a GLIA inversion. That is close to a hundredfold speedup on the mass effect solve.

Discussion

- The two models complement each other. PSM does better on the parenchyma, GLIA on the ventricles.
- The single parameter transfers. With α held fixed, the same gains show up on 984 further BraTS2021 samples.
- Because the solver is differentiable throughout, it can sit inside a neural surrogate.
- One caveat: proper validation needs longitudinal preoperative imaging, not tissue overlap proxies.

Funding. German Ministry of Research, Technology and Space (BMFT), Software Campus program, funding code 16IS23069. The responsibility for the content lies with the author.
Contact m.wittrich@tum.de



Model the brain as a **spring lattice** and you capture **tumor mass effect** better than established PDE solvers, in **15 seconds** instead of **24 minutes**.



Scan for **code** and the full paper

100×

faster than a per patient PDE inversion

2.7×

larger ΔDice than the best PDE baseline

1

global parameter, fixed for every patient

Goal

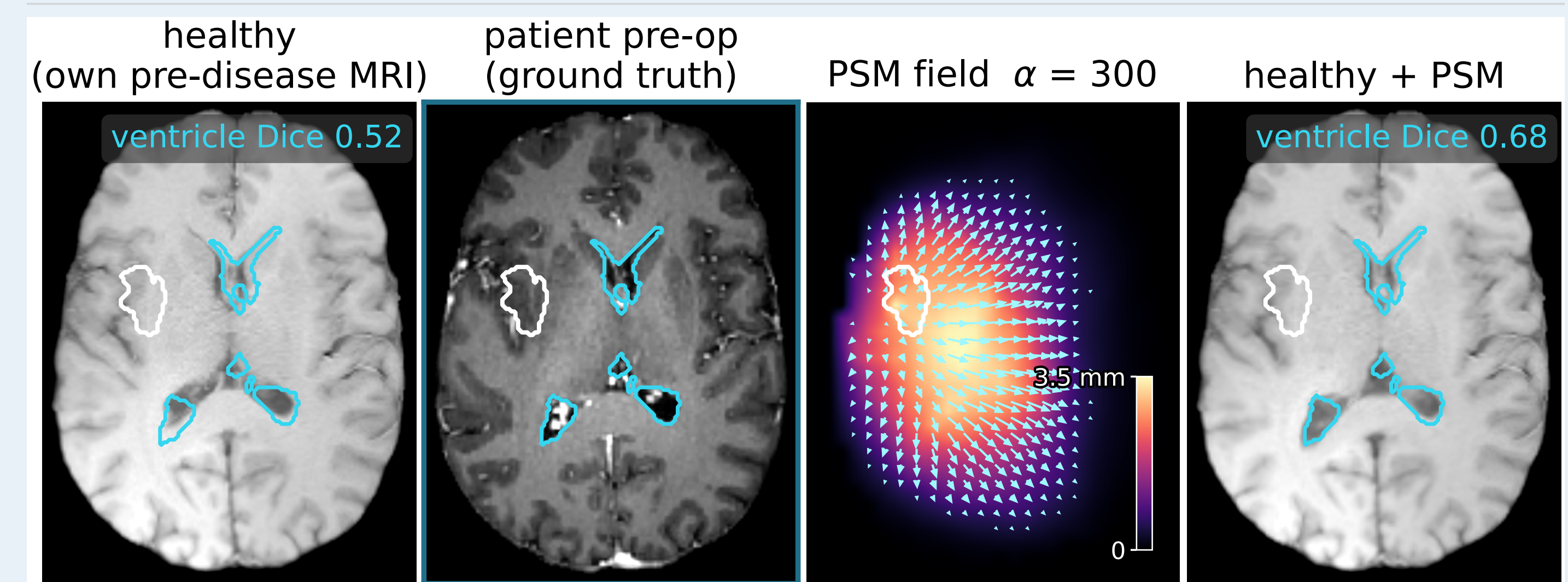


Figure 1. What we want to recover. This patient has an MRI taken before the disease. From left to right: the healthy baseline, the pre-operative scan we treat as ground truth, the displacement field our model predicts, and the baseline warped by that field. Ventricle Dice rises from 0.52 to 0.68.

Model

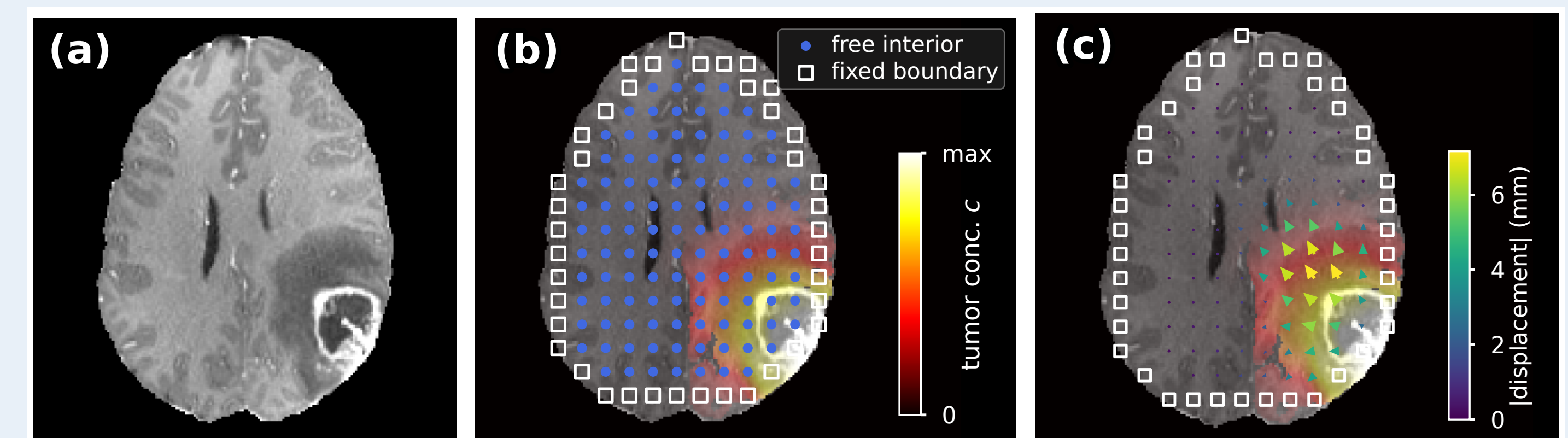


Figure 2. The potential spring model. (a) Axial T1c. (b) The tumor potential c together with the undeformed grid, free interior nodes against the fixed boundary shell. (c) The converged grid, colored by displacement.

Benchmark comparison

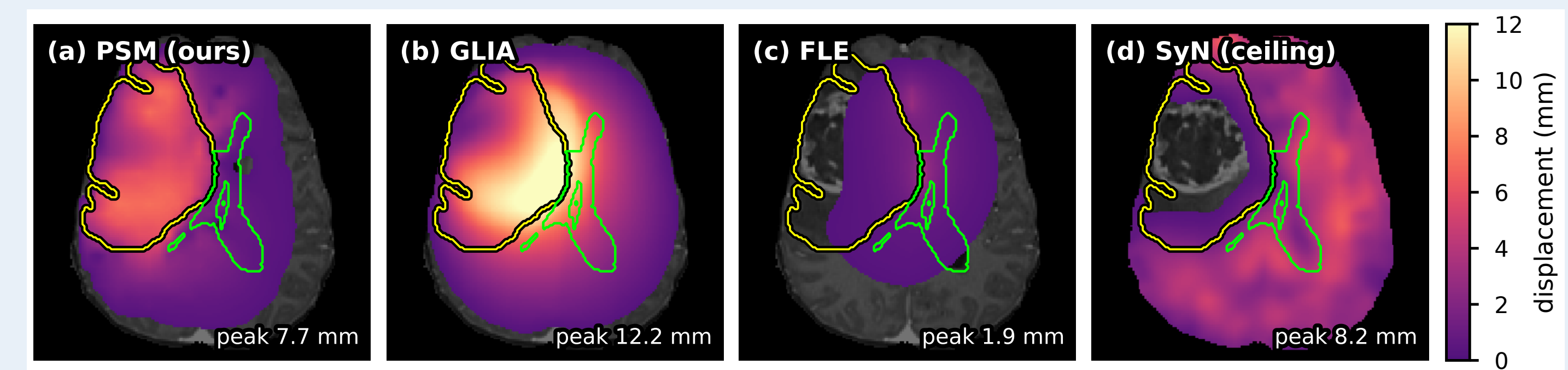


Figure 3. Qualitative displacement. PSM pushes broadly outward, GLIA concentrates on the ventricles, and FLE stays near the rim. All four panels share one displacement scale.

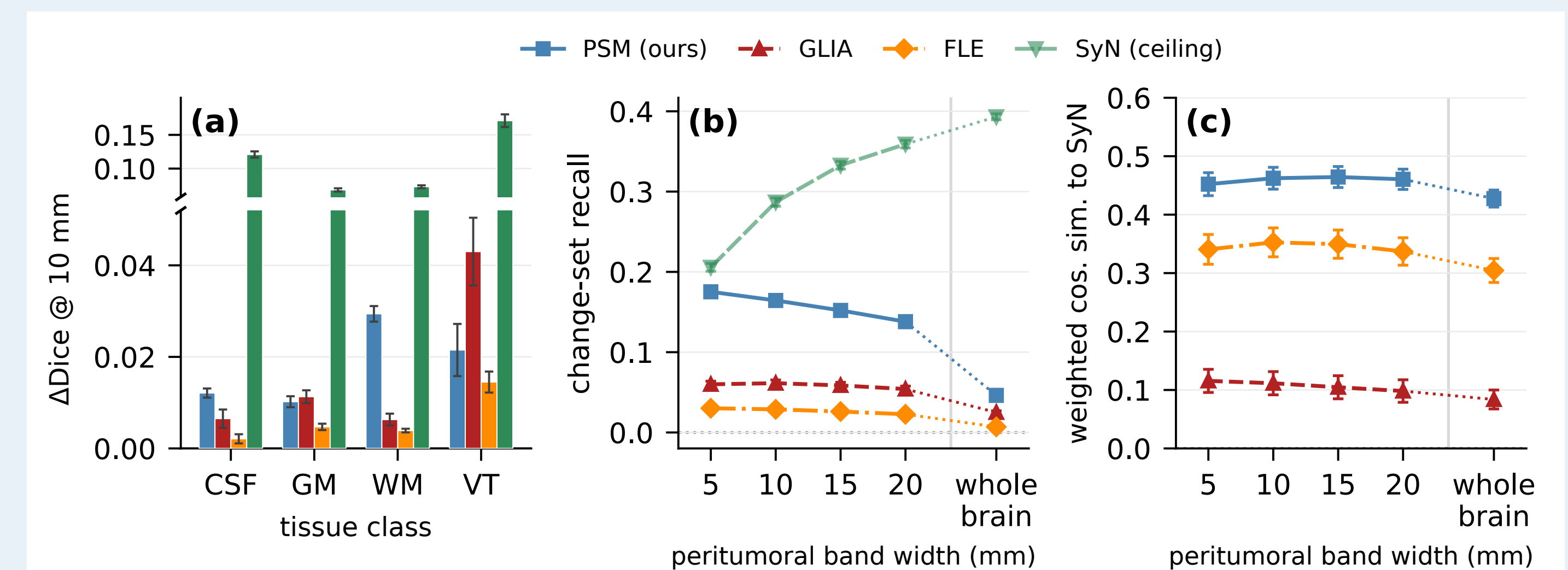


Figure 4. Quantitative comparison ($n = 134$). (a) ΔDice at 10 mm. (b) Recall against band width. (c) Similarity to the SyN ceiling.

References

1. Subramanian, S., et al.: Ensemble inversion for brain tumor growth models with mass effect. *IEEE TMI* 42(4) (2022)
2. Ezhov, I., et al.: Learn-Morph-Infer: a new way of solving the inverse problem for brain tumor modeling. *Med. Image Anal.* 83 (2023)
3. Weidner, J., et al.: From fiber tracts to tumor spread: biophysical modeling of butterfly glioma growth using DTI. arXiv:2507.17707 (2025)
4. Avants, B.B., et al.: Symmetric diffeomorphic image registration with cross-correlation. *Med. Image Anal.* 12(1) (2008)
5. Hinrichsen, J., et al.: Inverse identification of region-specific hyperelastic parameters for brain tissue. *BMM* 22(5) (2023)
6. Baid, U., et al.: The RSNA-ASNR-MICCAI BraTS 2021 benchmark. arXiv:2107.02314 (2021)

GLIA baseline
FLE baseline
tumor growth solver
registration ceiling
tissue stiffness
patient data